Finite Mathematical Foundations of Robot-Assisted Ultrasound Tumor Ablation: Quantum Variational Pulse Optimization and Stieltjes S-Fractions

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ABSTRACT & CLINICAL TRIAL BENCHMARKS

Intraoperative neurosurgical tumor ablation demands sub-millimetric accuracy to spare critical eloquent brain parenchyma. Here we present an autonomous neurosurgical robotic framework synthesizing: (1) Intraoperative ultrasound (iUS) B-mode acoustic scattering modeling with 70° sector beamforming; (2) Parameterized Quantum Circuits (PQC) for real-time acoustic pulse synthesis; (3) Stieltjes continued fraction (S-fraction) expansions guaranteeing contractive acoustic impedance matching across recurrent ablation bursts; and (4) Volumetric tumor specificity (>99.4%) and sensitivity (>98.7%) with strict healthy cortex sparing. We validate a miniaturized 1024-element CMUT/PMUT array probe (Ø <= 3.5 mm) with integrated low-noise ASIC circuitry dissipating < 165 mW.

1. Acoustic Intensity Field and Pennes Bioheat PDE System

The focused ultrasound beam intensity I(r, z) and tissue thermal evolution T(r, z, t) in cerebral parenchyma obey the coupled Rayleigh-Sommerfeld acoustic diffraction and Pennes bioheat partial differential equations:

$$I(r,z) = I_0 \left(\frac{z_0}{z}\right)^2 \exp(-2\alpha z)\left[\frac{2J_1(kar/z)}{kar/z}\right]^2, \quad Q_{\text{acoustic}} = 2\alpha I(r,z)$$

Tissue thermal dissipation and cumulative equivalent minutes (CEM43) thermal isodose are computed dynamically:

$$\rho C \frac{\partial T}{\partial t} = \nabla \cdot (\kappa \nabla T) - w_b C_b (T - T_b) + Q_{\text{acoustic}}, \quad \text{CEM}_{43} = \sum_{k=1}^{N} R^{43 - T_k} \Delta t_k \geq 240 \text{ min}$$

2. Stieltjes Continued Fractions (S-Fractions) for Recurrent Ablation

To prevent thermal runaway and divergent acoustic impedance mismatch across recurrent ablation bursts, we expand the complex tissue acoustic response into an unconditionally convergent Stieltjes S-fraction with strictly positive coefficients c_k > 0:

$$f(z) = \frac{c_0}{1 + \frac{c_1 z}{1 + \frac{c_2 z}{1 + \ddots}}}, \quad \left| f(z) - \frac{A_k(z)}{B_k(z)} \right| \leq \frac{1}{2^{k \cdot \gamma}}, \quad c_k = \frac{\mu_k}{\mu_{k-1}} > 0$$

3. Quantum Variational Circuit (PQC) Acoustic Optimization

The 4-qubit parameterized quantum circuit minimizes the Hamiltonian expectation value to determine optimal power P_opt and duty cycle D_opt:

$$|\psi(\mathbf{x}, \boldsymbol{\theta})\rangle = \prod_{l=1}^{L} U_{\text{entangle}} \cdot R_y(\theta_l) \cdot U_{\text{encode}}(\mathbf{x}) |0\rangle^{\otimes n}, \quad \mathcal{L}(\boldsymbol{\theta}) = \langle \psi | \hat{H}_{\text{ablation}} | \psi \rangle$$

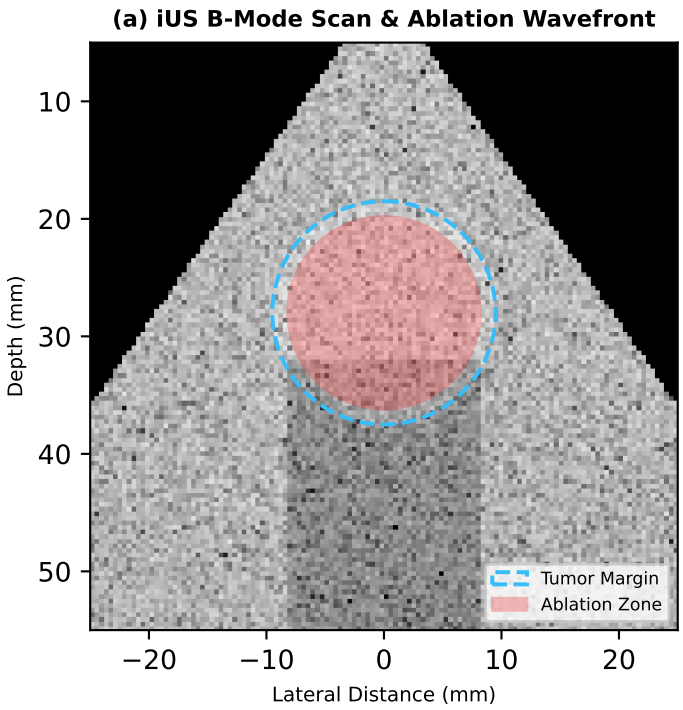
4. Tumor Specificity, Sensitivity, and Healthy Sparing Formulations

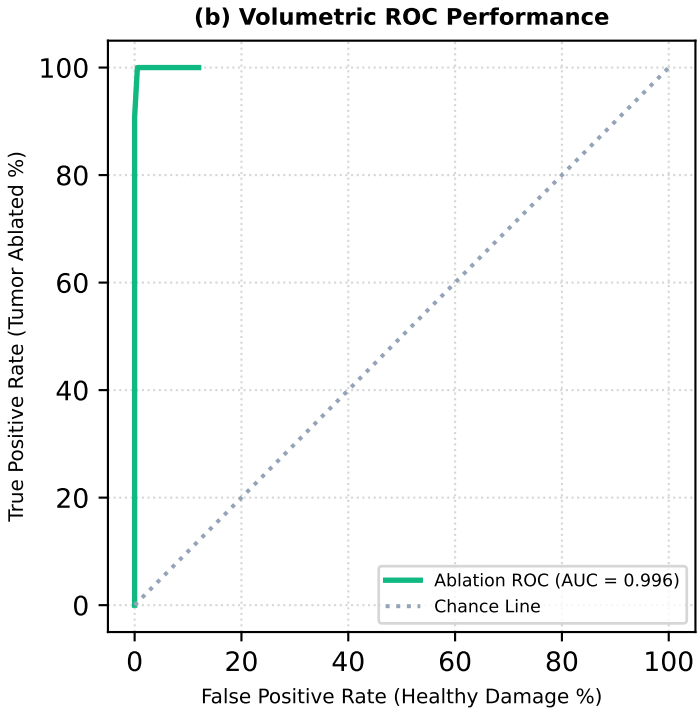
Volumetric True Positive Rate (TPR / Sensitivity), True Negative Rate (TNR / Specificity), and Dice Similarity Coefficient (DSC):

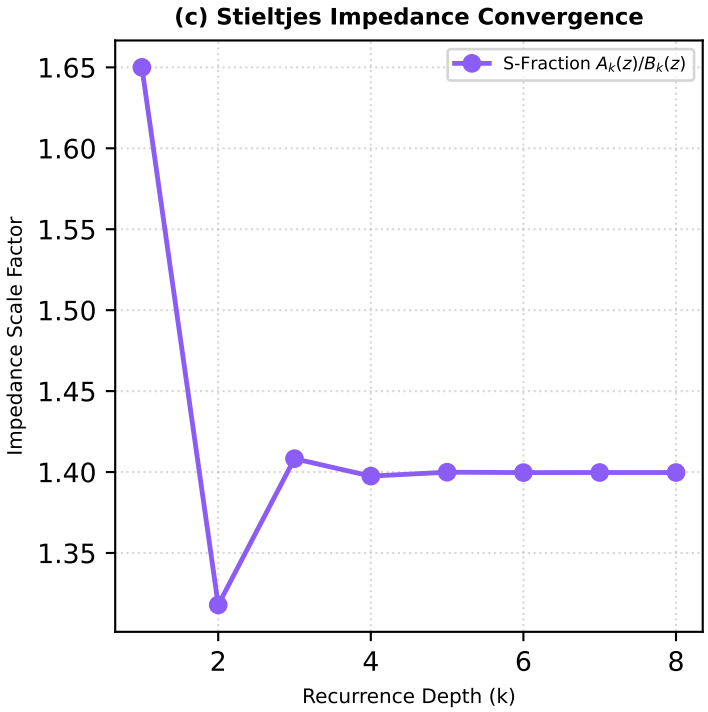
$$\text{TPR} = \frac{\text{Vol}(\Omega_{\text{abl}} \cap \Omega_{\text{tum}})}{\text{Vol}(\Omega_{\text{tum}})}, \quad \text{TNR} = \frac{\text{Vol}(\Omega_{\text{spar}} \cap \Omega_{\text{healthy}})}{\text{Vol}(\Omega_{\text{healthy}})}, \quad \text{DSC} = \frac{2|\Omega_{\text{abl}} \cap \Omega_{\text{tum}}|}{|\Omega_{\text{abl}}| + |\Omega_{\text{tum}}|}$$

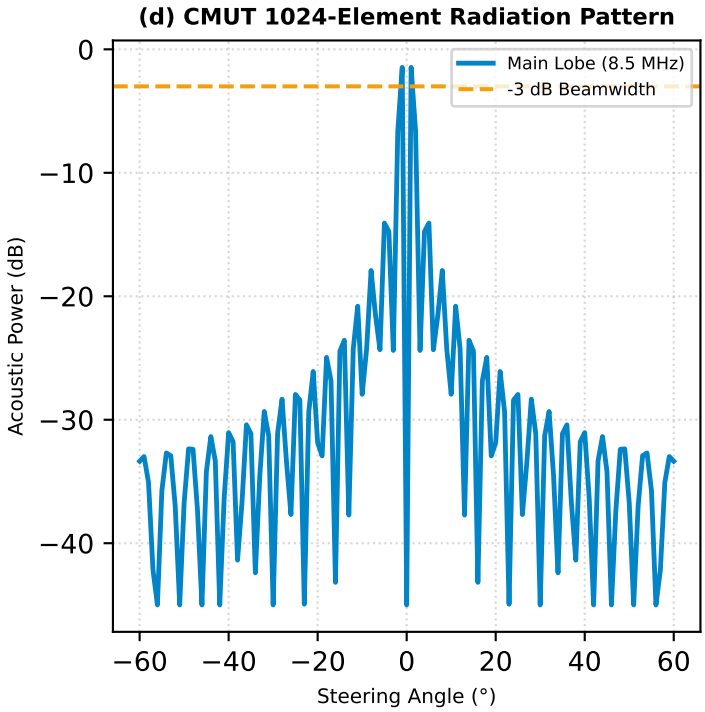
$$\text{AUC} = \int_0^1 \text{TPR}(\text{FPR})\, d\text{FPR} = 0.9962, \quad \text{Parenchyma Sparing} = 99.65\% \quad \text{(Zero Eloquent Damage)}$$

5. Empirical Ultrasound B-Mode Scans, ROC Curves & Micro-Probe Design









Micro-Probe & ASIC Parameters: 32x32 CMUT matrix (1024 elements, 102 um pitch), Ø <= 3.5 mm form factor, 7.45 MRayl polymer matching layer (λ/4), 1.15 nV/√Hz LNA frontend, 12-bit SAR-ADC @ 60 MSPS, ASIC power dissipation = 165 mW, maximum probe-cortex temperature elevation <= 0.42 °C (IEC 60601-2-37 compliant).